Universität Siegen

Naturwissenschaftlich-Technische Fakultät

4D and Quantum Vision Group

Quantum Paradigm Shifts in Combinatorial Optimization and Learning

Natacha Kuete Meli

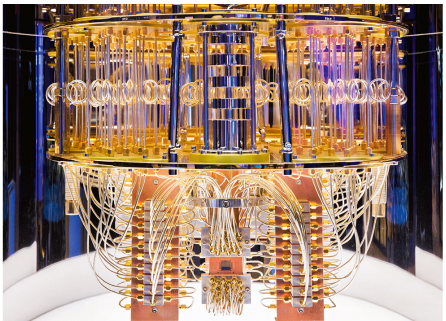
University of Siegen, Computer Vision Group

May 26, 2026

@Bielefeld University
Faculty of Technology

Background

Why Quantum Computing?



Expand computing hardware with an entirely new device type

- CPU/GPU/**QPU**/...
- Compute with laws of **quantum mechanics**.
- Unlock **new potentials**.

[1] *https://new.abb.com/news/detail/74736/quantum-computing-the-hype-and-hopes*

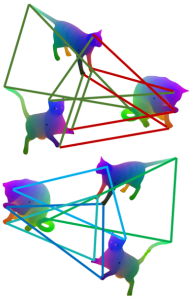
Why Quantum Computing?



Golyanik, 2020



Benkner, 2021



Birdal, 2021



Bhatia, 2023



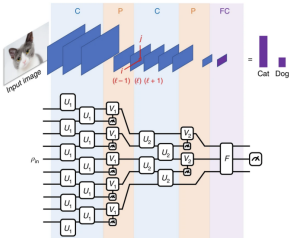
Kuete, 2025

And many more...

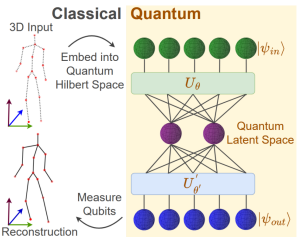
Optimization
& Energy minimization

- Quantum computer can solve some problems much faster than classical ones (Shor/Grover/...).
- Similar advantage possible in other applications?

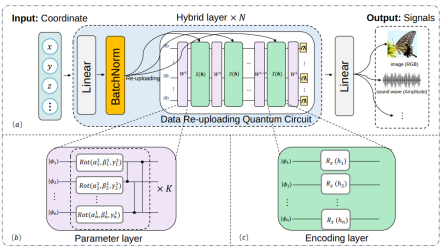
Why Quantum Computing?



Cong, 2019



Rathi, 2023



Zhao, 2024

And many more...

Quantum Machine Learning

- We can essentially use and train quantum computers like neural networks.

Quantum bit

Any particle that has two basis states and can be in a superposition of them.

Qubit

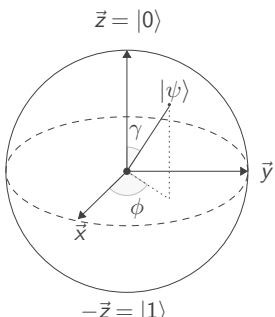
Let $|0\rangle$ and $|1\rangle$ be two basis vectors of $\mathbb{C}^2$, with

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A **qubit** is a one-particle system whose state vector is expressed as

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,,$$

with $\alpha, \beta \in \mathbb{C}$ subject to $|\alpha|^2 + |\beta|^2 = 1$.



Bloch sphere

Multi-qubit systems

The state of a n-qubit system is given by the **tensor (Kronecker)** product

$$|\psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle \in \mathbb{C}^{2^n}$$

of the n single-qubit states.

Basis states are correspondingly formed by tensor products of $|0\rangle$ and $|1\rangle$.
Example for $n = 3$:

$$|\psi\rangle = \alpha_{000}\,|000\rangle + \alpha_{001}\,|001\rangle + \cdots + \alpha_{111}\,|111\rangle$$

$|0\rangle \otimes |0\rangle \otimes |0\rangle$ or $\begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix}$ or 0 in binary basis

$|0\rangle \otimes |0\rangle \otimes |1\rangle$ or $\begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix}$ or 1 in binary basis

$|1\rangle \otimes |1\rangle \otimes |1\rangle$ or $\begin{pmatrix}0\\1\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix}$ or 7 in binary basis

Measurement in ~~a nutshell~~ the Bloch sphere

Recall

A qubit's state vector

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

collapses to the state $|0\rangle$ with probability to $|\alpha|^2$ and to the state $|1\rangle$ with probability $|\beta|^2$.

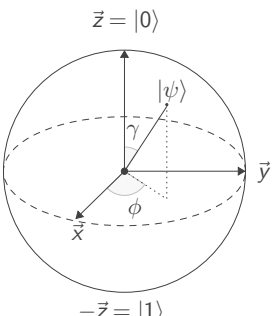
These are merely the results:

$$|\alpha|^2 = \langle\psi|0\rangle\,\langle 0|\psi\rangle \quad \text{and} \quad \beta^2 = \langle\psi|1\rangle\,\langle 1|\psi\rangle\,.$$

If we collect $|0\rangle\,\langle 0|$ and $|1\rangle\,\langle 1|$ in a matrix

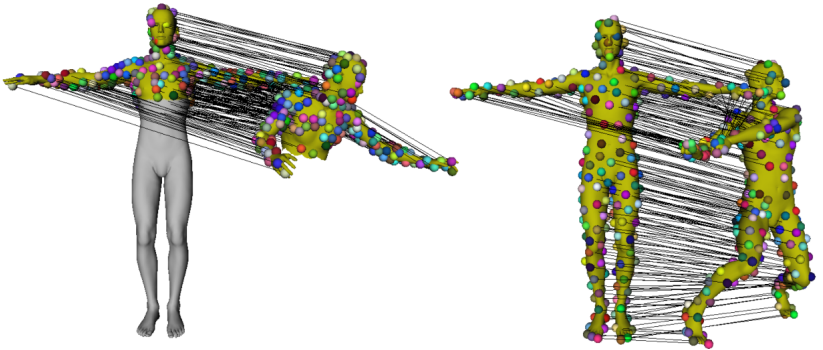
$$\sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

Then $\langle\psi|\sigma^z|\psi\rangle = |\alpha|^2 - |\beta|^2$ is the z-coordinate of $|\psi\rangle$ in $\mathbb{R}^3$.



Combinatorial Optimization

Quadratic Assignment Problems



QAP: Quadratic Assignment Problem

Find a **permutation matrix** P that minimizes

$$\min_{\mathbf{P}\in\{0,1\}^{n\times n}} \sum_{ijkl} \mathcal{D}(d_1(i,j),\, d_2(k,l))\, P_{ik}P_{jl} \qquad \text{s.t.} \qquad \sum_{i=1}^{n} P_{ij} = \sum_{j=1}^{n} P_{ij} = 1.$$

What are Quantum Computers Good At?

QUBO: Quadratic Unconstrained Binary Optimization [1]

Given $J \in \mathbb{R}^{k \times k}$, $h \in \mathbb{R}^k$, minimize

$$\min_{\mathbf{s} \in \{-1,1\}^k} \quad s^\top J s + h^\top s.$$

Problem Hamiltonian

Define a so-called **problem Hamiltonian**

$$H_P = \sum_{i,j=1}^{k} J_{ij} \sigma_i^z \sigma_j^z + \sum_{i=1}^{k} h_i \sigma_i^z,$$

$$\sigma_i^z = \ldots \otimes I \otimes \underbrace{\sigma^z}_{\text{pos. } i} \otimes I \otimes \ldots.$$

For a state

$$|q\rangle = \bigotimes_{i=1}^{k} |q_i\rangle\,, \quad |q_i\rangle = \begin{cases} |0\rangle\,, & \text{if} \quad s_i = 1 \\ |1\rangle\,, & \text{if} \quad s_i = -1, \end{cases}$$

$$\begin{aligned} s^\top J s + h^\top s &= \langle q | H_P | q \rangle \\ &= \text{energy of the quantum system in state } |q\rangle\,. \end{aligned}$$

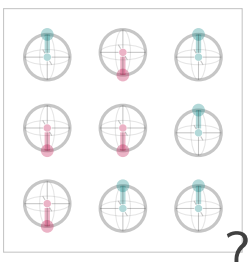
[1] *A. Das and B. Chakrabarti. Quantum annealing and related optimization methods. Springer Science & Business Media, 2005.*

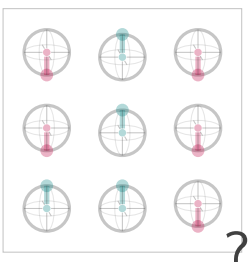
What are Quantum Computers Good At?

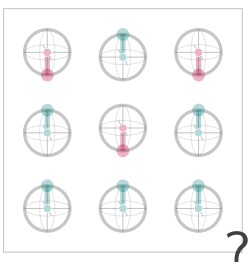
Given

$$H_P = \sum_{i,j=1}^{k} J_{ij}\sigma_i^z\sigma_j^z + \sum_{i=1}^{k} h_i\sigma_i^z,$$

what state minimizes the energy $\langle q|H_P|q\rangle$?







...

What are Quantum Computers Good At?

Adiabatic Quantum Computing [1]

A system evolving under the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

and **starting in the ground state of $H(0)$ remains in the ground state of $H(t)$** if the $H(t)$ changes slowly enough and the gap to the first excited state stays sufficiently large.

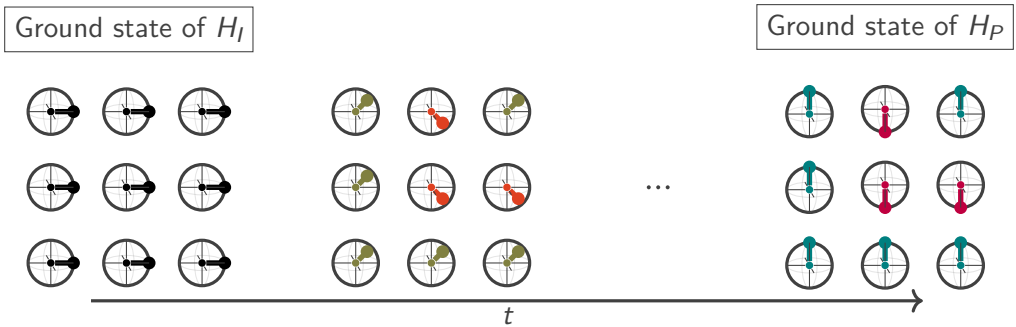
Define an initial H_I whose ground state is easy to prepare and let

$$H(t) = \left(1 - \frac{t}{T}\right) H_I + \frac{t}{T} H_P.$$

The adiabatic theorem brings the system in the ground state of H_P!!! ✔

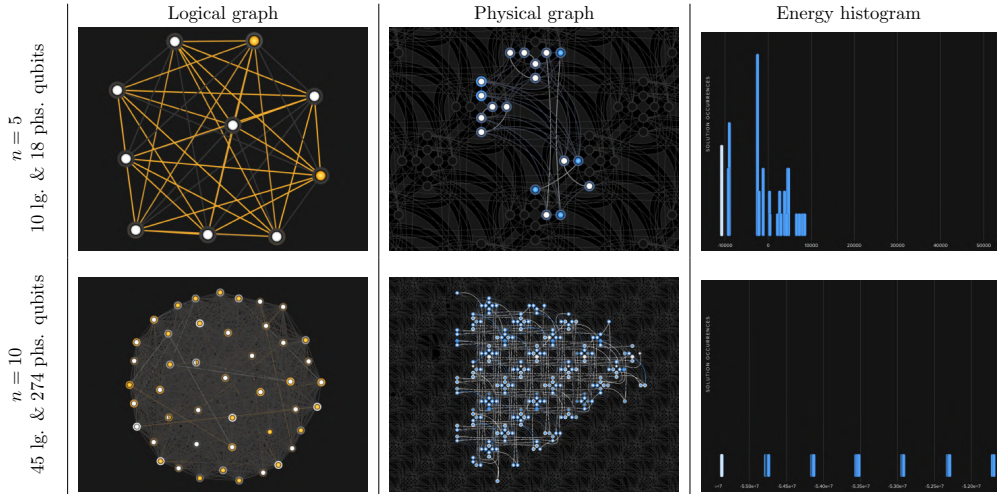
[1] *T. Albash and D. Lidar. Adiabatic quantum computation. Reviews of Modern Physics, 2018.*

What are Quantum Computers Good At?



$$H(t) = \left(1 - \tfrac{t}{T}\right) H_I + \tfrac{t}{T} H_P$$

What are Quantum Computers Good At?



[1] *Kuete Meli, et al. Qucoop: A versatile framework for solving composite and binary-parametrised problems on quantum annealers. CVPR, 2025.*

QAP vs. QUBO

What quantum computers are good at

$$\min_{\mathbf{s}\in\{-1,1\}^k} \quad s^\top Js + h^\top s \quad \overset{\text{Substitution: } s=2x-1}{\longrightarrow} \quad \min_{\mathbf{x}\in\{0,1\}^k} \quad x^\top Qx$$

What we want

$$\min_{\mathbf{P}\in\{0,1\}^{n\times n}} \quad \sum_{ijkl} \mathcal{D}(d_1(i,j),\, d_2(k,l))\; P_{ik}P_{jl} \qquad \text{s.t.} \qquad \sum_{i=1}^{n} P_{ij} = \sum_{j=1}^{n} P_{ij} = 1$$

$$\Updownarrow$$

$$\min_{p\in\{0,1\}^{n^2} | Ap=b} \quad p^\top Qp$$

QAP to QUBO

Traditional QUBO mapping

$$\min_{p \in \{0,1\}^{n^2}} p^\top Qp + \lambda \|Ap - b\|_2^2$$

- n^2 variables
  $\Rightarrow$ computationally heavy.
- Choosing λ is tricky
  $\Rightarrow$ feasibility vs. data fidelity.
- Solved $n \leq 5$ on current devices.

Better, parameterize P:

$$(c, d, b, a) \to (\mathbf{a}, d, b, c) \to (\mathbf{a}, \mathbf{b}, d, c) \to (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d})$$

Fix position 1 — Transpose (1, 4)
Fix position 2 — Transpose (2, 3)
Fix position 3 — Transpose (3, 4)
Done

$$\mathbf{P(x)} = (\mathbf{3}, \mathbf{4})^{\mathbf{1}} \cdot (\mathbf{2}, \mathbf{4})^{\mathbf{0}} \cdot (\mathbf{2}, \mathbf{3})^{\mathbf{1}} \cdot (\mathbf{1}, \mathbf{4})^{\mathbf{1}} \cdot (\mathbf{1}, \mathbf{3})^{\mathbf{0}} \cdot (\mathbf{1}, \mathbf{2})^{\mathbf{0}}$$
$$= \prod_{\mathbf{i}=1}^{\mathbf{k}} \mathbf{T_i^{x_i}}$$

Binary-Parametrised Problem

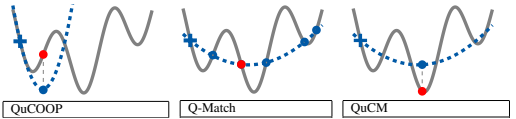
$$\min_{x \in \{0,1\}^k} p^\top(x) Qp(x)$$

QUBO only if $p(x)$ linear in x.
Iteratively linearize $p(x) \Rightarrow$ **sequel of QUBOs!**

QAP to QUBO

Linearize: $P(x) = \prod_{i=1}^{k} \mathbf{T}_i^{x_i}$

$$= I + \sum_{i=1}^{k} x_i (T_i - I) + \sum_{\substack{i,j=1 \\ i<j}}^{k} x_i x_j (T_i - I)(T_j - I) + \cdots + \prod_{i=1}^{k} x_i (T_I - I)$$



QuCOOP[Kuete Meli et al, 2025]:

$$p(x) = p^{(t)} + \nabla_x p^{(t)} \left(x - x^{(t)} \right)$$

$$\min_{x \in \{0,1\}^k} p^\top(x) Q p(x) + \lambda \| p(x) \|_2^2$$

Q-Match[Benkner et al, 2021]:

$$P(x) = \underbrace{\left(\prod_{i=1 \mid T_i \text{ disjoint}}^{k} T_i^{x_i} \right)}_{I + \sum_i x_i (T_i - I)} P^{(t)}$$

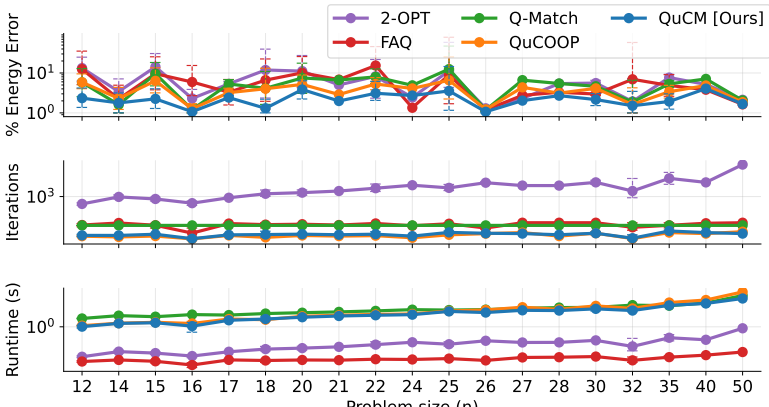
$$\min_{x \in \{0,1\}^k} p^\top(x) Q p(x)$$

QuCM[under review]:

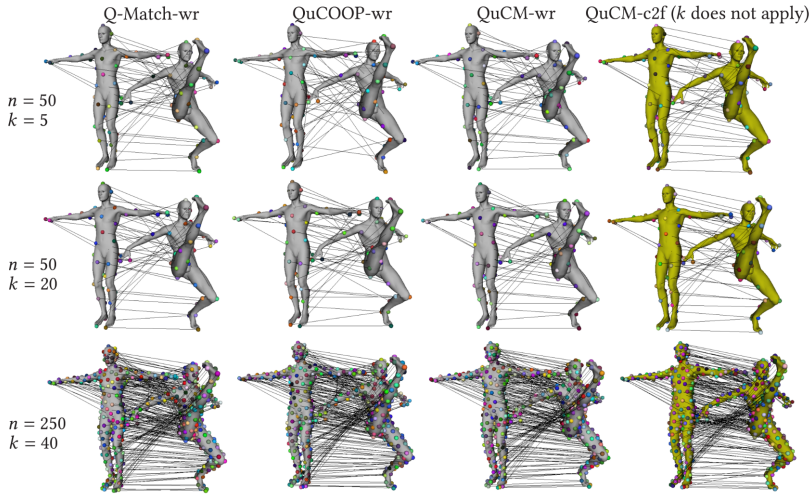
$$p(x) = \left(I + \sum_i x_i (T_i - I) \right) P^{(t)}$$

$$\min_{x \in \{0,1\}^k} p^\top(x) Q p(x) + \lambda \| p(x) \|_2^2$$

QAP Solvers on QAPLIB



QAP Solvers on Shape Matching



Interference-based Learning

What is Supervised Learning?

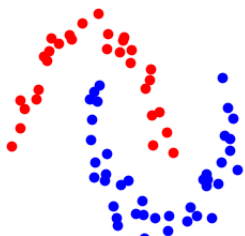
Given is a **labeled training dataset** $\{(x^k, y^k)\}_{k=0}^{n}$, $x^k \in \mathbb{R}^2$, $y^k \in \{-1, 1\}$.
Now given a new, unseen sample x_{new}, what is its label y_{new}?

Minimize a cost function

$$\min_{\theta} \sum_{\text{training sample } x^k} \mathcal{L}\Big(y^k,\ f(x^k, \theta)\Big)$$

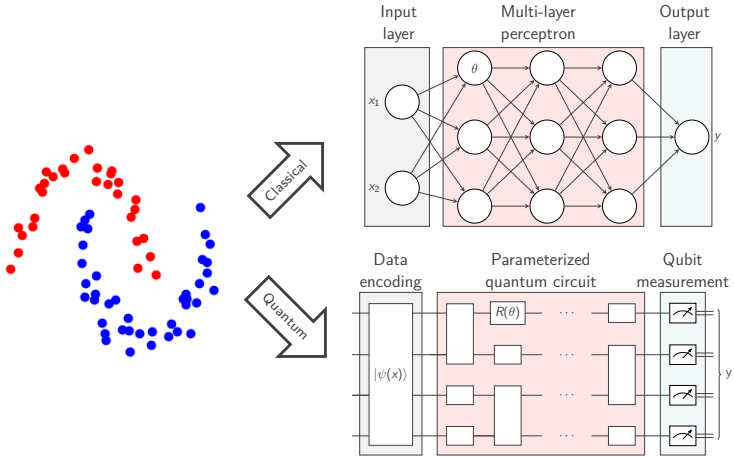
Loss
e.g. $\|y_i - f(\ldots)\|^2$

Parameterized function that depends on θ



- Model and parameterize f;
- Use the trained function f to predict y^{new}.

SOTA QML Paradigm



Parameter shift-rule [1]:

$$\frac{\partial}{\partial\theta} f = \frac{1}{2}\left(f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})\right)$$

- Computationally heavy
- Barren plateau

[1] *Schuld et al. Evaluating analytic gradients on quantum hardware. Physical Review A, 2019.*

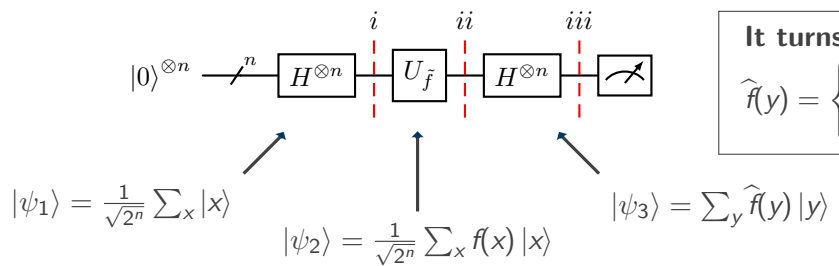
What are Quantum Computers Good At?

Bernstein-Vazirani [1]

Given observations $(x, f(x))$ with the assumption

$$f(x) = s^\top x \bmod 2,$$

for $x, s \in \{0,1\}^n$, recover the *unknown* s.



It turns out that:

$$\widehat{f}(y) = \begin{cases} 1, & \text{if } y = s, \\ 0, & \text{otherwise.} \end{cases}$$

[1] *E. Bernstein and U. Vazirani. Quantum complexity theory. ACM symposium on Theory of computing, 1993.*

Bernstein-Vazirani vs. Learning

What quantum computers are good at

Given observations $(x, f(x))$ with the assumption

$$f(x) = s^\top x \bmod 2,$$

for $x, s \in \{0,1\}^n$, recover the *unknown* s.

What we want

Function f does not necessarily fulfill

$$f(x) = s^\top x \bmod 2.$$

- $(x, f(x))$: Training samples
- $f(x) = s^\top x \bmod 2$: Model, architecture
- s: Model parameters

What are Quantum Computers Good At?

Theorem (Fourier expansion on $\mathbb{Z}_2^n$, [O'Donnell, 2014])

Any function $f\colon \{0,1\}^n \to \mathbb{R}$ admits a unique expansion

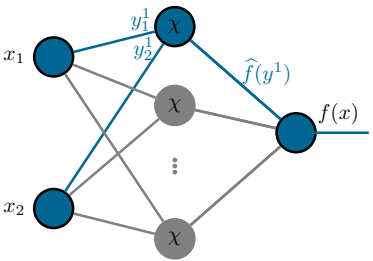
$$f(x) = \sum_{y\in\{0,1\}^n} \widehat{f}(y)\,\chi_y(x), \quad \chi_y(x) = (-1)^{y^\top x}$$

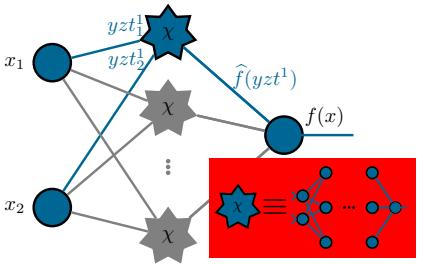
called the Fourier expansion of f, and $\widehat{f}(y) \in \mathbb{R}$ are Fourier coefficients in the basis χ.

From $|\psi_3\rangle = \sum_y \widehat{f}(y)\,|y\rangle$:

- Sample y with prob. $|\widehat{f}(y)|^2$
- Knowing the expression of χ, recover $\widehat{f}(y) = \sum_x f(x)\chi_y(x)$ from training data.
- **Efficient approximation:** $f(x) = \sum_{y \text{ sampled}} \widehat{f}(y)\chi_y(x)$.

Bernstein Vazirani Networks (BVNs)





Standard:

$$f(x) = \sum_y \widehat{f}(y)\,\chi_y(x)$$

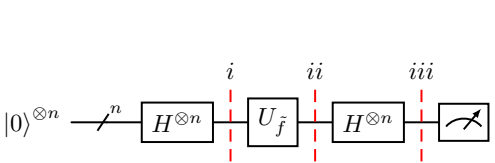
$$\chi_y(x) = (-1)^{y^\top x}$$

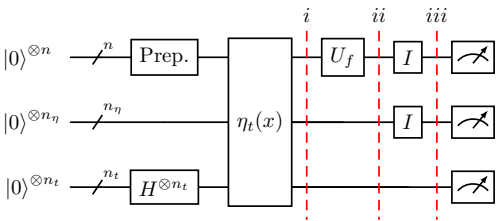
Generalized:

$$f(x) \approx \sum_{yzt} \widehat{f}(y)\,\chi_{yzt}(x)$$

$\chi_{yzt}(x) =$ **Something more expressive**

Bernstein Vazirani Networks (BVNs)





Standard:

$$f(x) = \sum_y \widehat{f}(y)\, \chi_y(x)$$

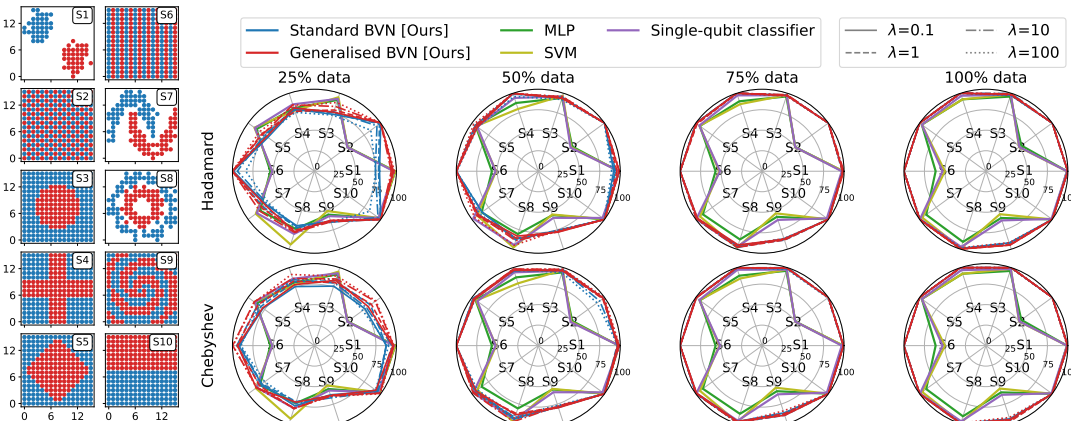
$$\chi_y(x) = (-1)^{y^\top x}$$

Generalized:

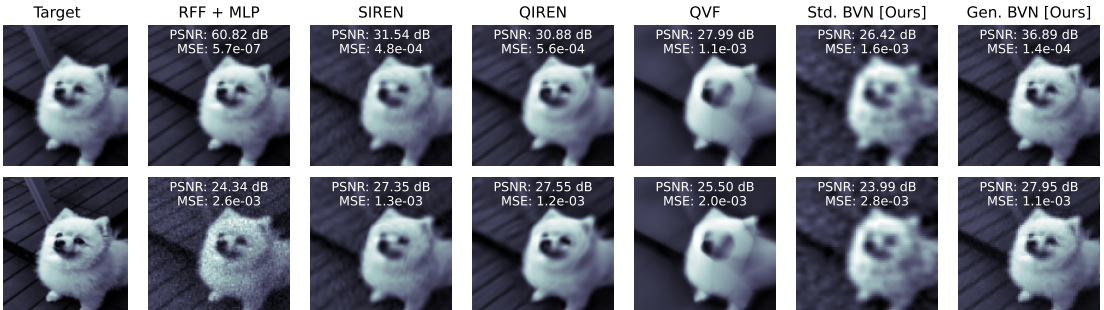
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$\chi_{yzt}(x) =$ **Something more expressive.**

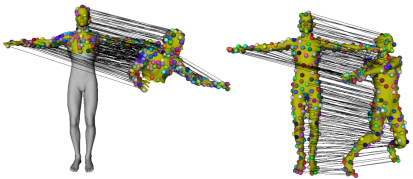
BVNs on Binary Classification

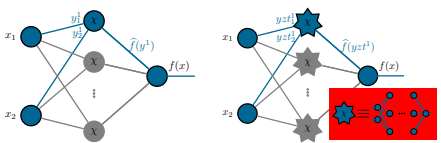


BVNs on Implicit Image Representation



Summary





Combinatorial Optimization:

- AQC is a good heuristic for optimization.
- Iterate & refine to scale up the applications.
- Often used in hybrid quantum-classical loops.

Interference-based Learning:

- Interference as a principled learning paradigm.
- Much more efficient that variational models.
- Still needs to be validated on broader tasks.

Thanks! Questions?

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